Modeling and Validation of a Soccer Ball's Three-Dimensional Flight to the Goal

Problem B

Abstract

Gravity, drag and Magnus lift govern a soccer ball's three-dimensional flight. We formulate a constant-spin model and compare gravity alone, constant drag, constant drag and lift, and complete variable coefficients. Empirical aerodynamic relations and a matched wind-tunnel comparison provide the coefficient basis.

Five prescribed kick situations give six initial states and 24 forward flights, illustrating the roles of speed, elevation and spin direction. Three actual match penalties provide 39 original sampled frames. After conversion to penalty-mark coordinates, three selected positions per event enter 12 bounded inversions, whose continuations are compared with later measured samples. Results report fitting residuals, held-out errors, crossings and ground termination individually. A separate same-state curved-shot calculation quantifies angular damping, shifting the crossing by 0.0343 m. Sensitivity and crossing-distribution analysis remain reserved for Section 5.

Keywords: Soccer ball, aerodynamic drag, Magnus effect, inverse prediction, trajectory comparison

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1 Introduction

1.1 Problem Background

A penalty kick places stringent spatial and temporal demands on the goalkeeper. The goalkeeper must reach an appropriate location at the right time, while deciding when and in which direction to move. Studies of penalty anticipation discuss this timing problem and show that shot speed and kicking technique affect the predictability of shot direction [1, 2]. Such work also considers cues from the kicker before contact; the present problem instead focuses on predicting the ball's later position from its measured positions after launch.

The path of a kicked soccer ball is not generally a straight line. Gravity bends the trajectory downwards, air resistance slows the ball, and rotation can produce a sideways or vertical aerodynamic force. Bray and Kerwin combined aerodynamic forces with equations of motion and compared the resulting trajectories with experimental free kicks [5]. Myers and Mitchell analyzed three-dimensional spinning-ball motion and the influence of drag and spin [14]; Maponi and Piombin developed a numerical framework that also accommodates wind [12].

The aerodynamic behavior is also affected by the ball's surface. Wind-tunnel measurements by Asai et al. examined the relationship between flow conditions and soccer-ball forces [15], while Paudel and Yan's simulations compared a football with realistic seam geometry against a smooth sphere [13]. These studies motivate a spherical geometric approximation together with an examination of how aerodynamic coefficients affect the predicted flight.

Measurements of real flights provide a basis for evaluating such models. Griffiths et al. tracked the three-dimensional motion and spin of a marked football in field conditions [3]. Goff and Carré used camera-recorded trajectories to estimate aerodynamic coefficients [19]; subsequent work by Goff et al. combined measurements from 29 flights to construct speed-dependent drag and lift curves [4].

In this problem, the initial position and two early positions are given, whereas the initial velocity and spin are not directly specified. We therefore use the equations of motion to find states compatible with the early observations and calculate their goal-plane crossings. The DFL open dataset combines match events and optical tracking data for professional matches [10, 11], allowing predicted crossings to be compared with later recorded positions.

1.2 Problem Restatement

We study a ball travelling from the penalty mark to a goal 7.32 m wide and 2.44 m high, 11 m away. The motion and prediction problem can be organized into three stages:

1. **Ball–foot contact.** Contact establishes launch velocity and spin. These initial conditions determine the forward examples and the unknown quantities reconstructed in Section 4.

2. **Airborne flight.** Gravity, drag and Magnus lift develop the trajectory according to the governing equations established in Section 3.

3. **Goal arrival and shot outcome.** The goal-plane crossing connects the flight to the shot result. Variation in initial motion, aerodynamic parameters and environmental conditions can produce a distribution of crossing locations; Section 5 is reserved for that subsequent sensitivity and uncertainty analysis.

We first compare prescribed kicks, then estimate initial states from three real penalties and compare predictions with later recorded positions. Figure 1 summarizes this physical and computational sequence.

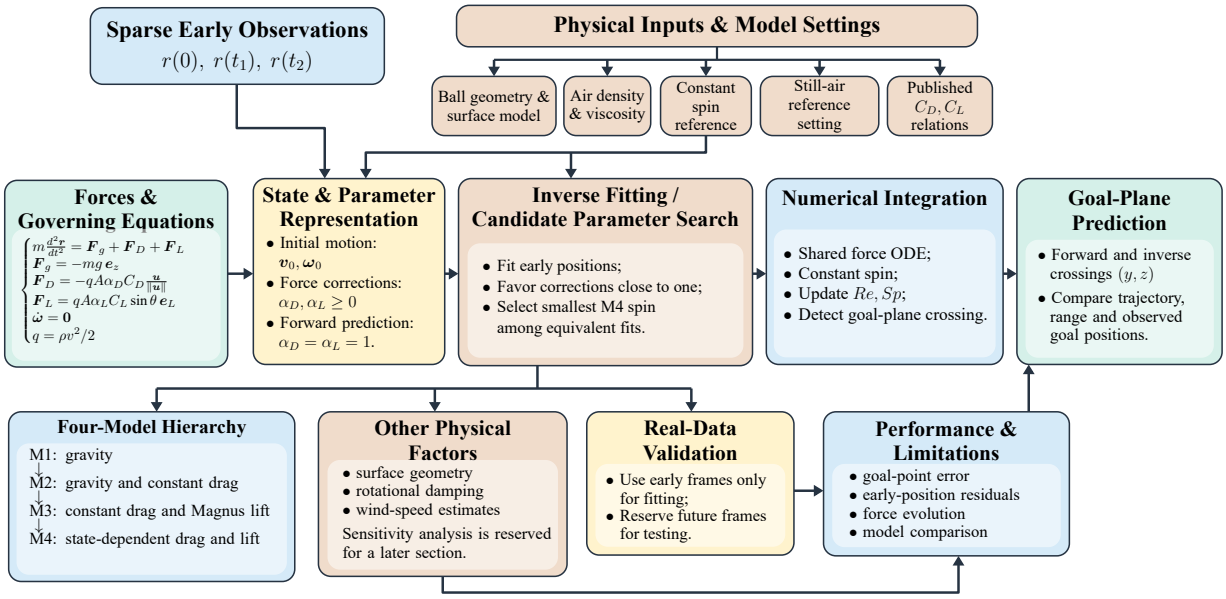


Figure 1: Three-force modeling, prescribed forward scenarios and real-penalty inverse comparison.

2 Preparation of the Models

2.1 Assumptions

- **Contact and model start.** Prescribed examples begin just after contact. Real inversions define $t = 0$ at the first available tracking frame; contact is represented through the initial motion.
- **Rigid spherical ball.** The ball is idealized as a rigid sphere with constant mass, radius and cross-sectional area $A = \pi R^2$; its centre of mass coincides with its geometric centre [12, 5, 14]. Empirical football coefficients represent the average aerodynamic effect of its surface. Section 6.1 discusses seams and asymmetric geometry.
- **Free-flight motion.** Gravity, aerodynamic drag and spin-induced lift act during free flight [12, 5]. The flight considered ends at the first contact with the ground, goal structure or another object; collisions, bouncing and rolling are excluded. Integration also stops at the selected goal plane.
- **Uniform gravity.** Gravitational acceleration is constant, with $g = 9.81\,\mathrm{m/s^2}$.
- **Atmospheric properties.** Air density and dynamic viscosity are spatially uniform during each flight. Aerodynamic forces are represented using empirical drag and lift coefficients.

- **Wind.** The main calculation uses still air. When wind is included, its velocity vector is spatially and temporally constant during each flight; small-scale turbulence and rapidly varying gusts are neglected. Section 6.3 examines how different wind conditions affect relative airflow and force magnitude.
- **Spin during flight.** The main model neglects aerodynamic torque, so its spin vector remains constant. Section 6.2 examines this approximation by adding rotational damping to the same initial state.
- **Observations and unknowns.** Three specified real-penalty positions enter a bounded deterministic fit, and subsequent positions supply the comparison. Released Z values are interpreted as metres and centre height. Section 4 states the search ranges for velocity, spin and force corrections; Section 5 is reserved for their effect on crossing distributions.

2.2 Notations

Table 1: Principal quantities and competition reference values.

Symbol	Meaning	Value or unit
$\boldsymbol{r}, \boldsymbol{v}$	Ball-centre position and velocity	m; m/s
$\boldsymbol{\omega}, \omega$	Spin vector and its magnitude	rad/s
m, R, A	Mass, radius and frontal area	0.43 kg; 0.11 m; πR^2
g	Gravitational acceleration	9.81 m/s^2
$\boldsymbol{u}, v$	Air-relative velocity and its magnitude	m/s
ρ	Air density	1.2 kg/m^3
$\nu, \mu = \rho\nu$	Kinematic and dynamic viscosity	1.50×10^{-5} m^2/s; Pa s
Re, Sp	Reynolds number and spin parameter	dimensionless
C_D, C_L	Drag and transverse lift coefficients	dimensionless
α_D, α_L	Multiplicative force corrections	dimensionless
$\theta, \boldsymbol{e}_L$	Spin–airflow angle and Magnus direction	rad; unit vector
X_g	Local goal-plane coordinate	11 m
$\boldsymbol{Q}, I_b$	Damping torque and moment of inertia (Section 6)	N m; kg m^2

The competition mass, radius and gravity follow the reference values of Maponi and Piombin [12]; the air density follows Asai et al. [15]. The viscosity is a declared reference air setting, with $\mu = \rho\nu$. The three match penalties use the same declared reference settings.

3 Model Establishment

The same Newtonian equations generate all four models by selecting the included aerodynamic terms. We first describe the geometry, then connect the forces to the surrounding flow and ball rotation.

3.1 Geometry and Goal Plane

The local horizontal origin is the nominal penalty mark, with x towards the goal, y lateral and z upward. The goal plane is $x = 11$ m and the opening is $-3.66 \le y \le 3.66$ m, $0 \le z \le 2.44$ m. Section 4.2 converts the measured pitch-centred coordinates.

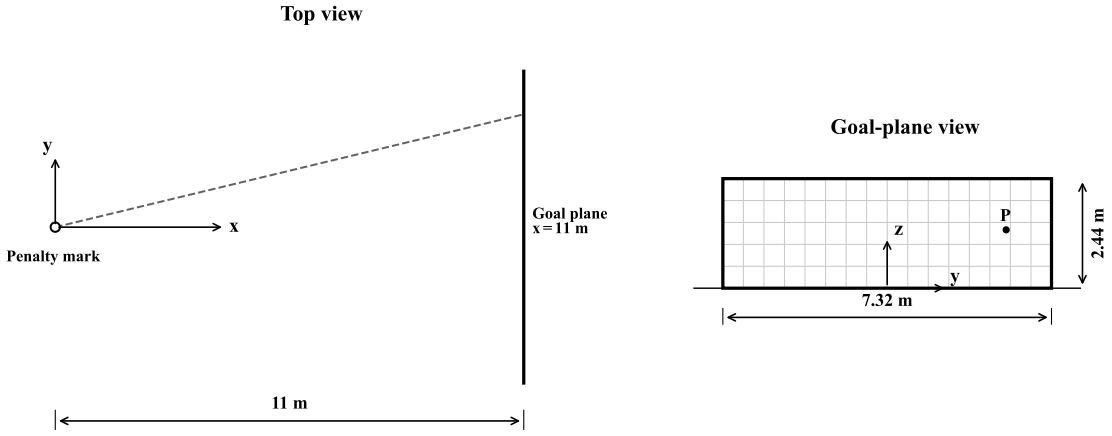


Figure 2: Penalty-kick geometry in local coordinates. The dashed line and point P indicate an illustrative crossing.

A clean opening crossing requires $|y| \le 3.66 - R$ and $R \le z \le 2.44 - R$. Descending centre height $z = R$ defines ground contact. Every model uses these same event definitions.

3.2 Force Analysis

3.2.1 Gravity

Gravity supplies a constant downward force,

$$\boldsymbol{F}_g = -mg\boldsymbol{e}_z, \tag{1}$$

where $\boldsymbol{e}_z$ is the upward vertical unit vector. It produces downward curvature even when all aerodynamic forces are removed.

3.2.2 Aerodynamic Drag

A moving ball transfers momentum to the air in its wake. The mass flow through area A scales as $\rho A v$, and the velocity change scales as v, giving a force scale $\rho A v^2$. Introducing dynamic pressure $q = \rho v^2/2$ gives the conventional drag expression [19, 16],

$$\boldsymbol{F}_D = -\tfrac{1}{2}\rho A \alpha_D C_D(Re, Sp) v \boldsymbol{u}, \qquad \boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}, \quad v = \|\boldsymbol{u}\|. \tag{2}$$

Its quantities have the following physical roles:

1. **Air density $\boldsymbol{\rho}$.** The density measures air mass per unit volume. We use the uniform reference value 1.2 kg/m^3 reported by Asai et al. [15].

2. **Projected area $A = \pi R^2$.** A larger radius intercepts more airflow; the spherical geometry fixes this area throughout flight.

3. **Relative speed and direction v, u.** The quadratic speed dependence gives stronger resistance at higher speed, and the negative direction opposes airflow. The still-air main model has $u = v$.

4. **Drag coefficient $C_D(Re, Sp)$.** The coefficient represents boundary-layer transition, separation and wake formation. Following Asai et al. and Kiratidis and Leinweber [15, 16], we define

$$Re = \frac{2Rv}{\nu}, \qquad Sp = \frac{R\omega}{v}, \qquad \omega = \|\boldsymbol{\omega}\|, \quad \nu = \frac{\mu}{\rho}. \tag{3}$$

Re compares inertial and viscous flow effects; Sp compares rotating surface speed with airspeed. Constant spin makes Sp rise as the ball slows.

The drag crisis is the rapid reduction of drag associated with boundary-layer transition. A useful coefficient law should describe this transition as well as the slower variation in the supercritical regime. According to Kiratidis and Leinweber [16], their Eq. (5) represents zero-spin drag by

$$\begin{aligned} f(v) = {} & \frac{a - b_{\text{min}}}{1 + \exp[(v - v_c)/v_s]} + b_{\text{min}} \\ & + \frac{v - v_{\text{min}}}{1 + \exp[-(v - v_{\text{min}})/v_s]} \frac{b_{\text{max}} - b_{\text{min}}}{v_{\text{max}} - v_{\text{min}}}. \end{aligned} \tag{4}$$

The coefficient a sets the low-speed level, v_c locates the transition, and v_s controls its width. The remaining parameters set the level and slope of the high-speed tail. Table 2 gives the Teamgeist parameters reported in their Table 1 [16].

Table 2: Teamgeist zero-spin drag parameters from Kiratidis and Leinweber [16].

a	v_c	v_s	b_{min}	b_{max}	v_{min}	v_{max}
—	m/s	m/s	—	—	m/s	m/s
0.4927	12.58	1.071	0.1440	0.1540	23.17	35.00

For a spinning Teamgeist, Kiratidis and Leinweber's Eq. (7) adds a linear spin correction with slope 0.514, fitted to the supercritical wind-tunnel data [16]:

$$C_D(Re, Sp) = f\left(\frac{Re\nu}{2R}\right) + 0.514Sp. \tag{5}$$

Thus drag changes with speed even at zero spin, and also increases with rotation. The Teamgeist parameter set is used throughout the coefficient comparison; other ball surfaces require their own parameters.

The prescribed launch speeds are 14–27 m/s, giving $Re = (2.05\text{–}3.96) \times 10^5$ within the speed range represented by the Teamgeist fit. The relation describes transition and supercritical flow up to the reference 35 m/s [16]. The comparison figures below use measured $Re = 455172$; inverse flights update the coefficients at their actual Re and Sp.

5. **Drag correction $\alpha_D \geq 0$.** This dimensionless multiplier equals one in forward calculations and is estimated from early positions in inverse calculations.

Comparison with published wind-tunnel data. Asai et al. [15] measured soccer-ball forces in Tokai University's low-speed recirculating wind tunnel using a six-component balance (1.5×1 m test section; about 1% turbulence). We use their spinning full-size Teamgeist FRP replica: $R = 0.11$ m, $\rho = 1.2$ kg/m^3, airspeed 22–30 m/s and spin 2–10 rev/s. Five points per coefficient are digitized from their Figures 8 and 9 at $Re = 455172$ (about 30 m/s); the reported side force C_S represents transverse Magnus lift here. The speed–Re correspondence gives $\nu \simeq 1.45 \times 10^{-5}$ m^2/s for this matched comparison.

Figure 3 reproduces the increase of drag with Sp: the coefficient MAE is 0.00985 and RMSE is 0.01127. Both curves use the same experimental condition and measured spin interval.

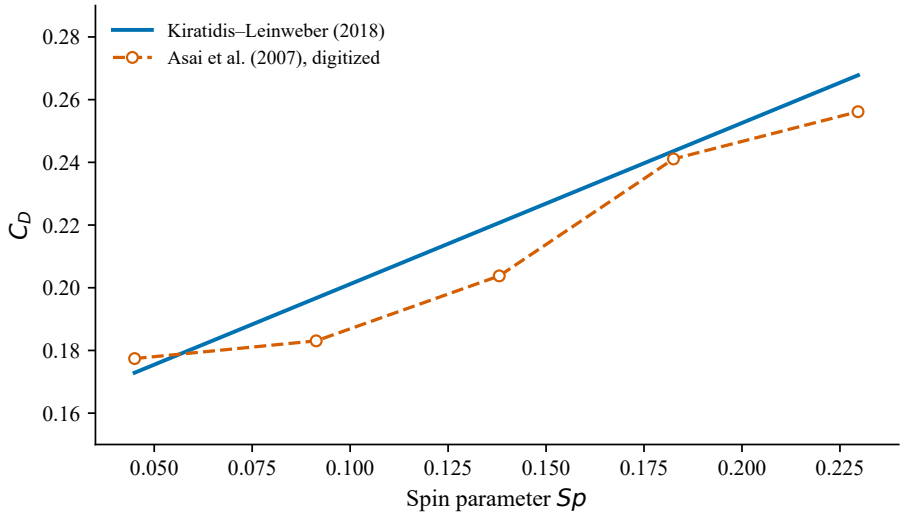


Figure 3: Teamgeist drag at $Re = 455172$: the complete empirical relation [16] and five digitized Asai measurements [15], Figure 8.

3.2.3 Magnus Lift

Rotation changes the flow on opposite sides of the ball and creates a transverse pressure force. Using the drag dynamic-pressure scale gives $F_L = qAC_L$ for transverse spin [19, 16]. For a three-dimensional spin axis, we adopt

$$\begin{aligned} \boldsymbol{F}_L &= \tfrac{1}{2}\rho A \alpha_L C_L(Re, Sp) v^2 \sin\theta \, \boldsymbol{e}_L, \\ \boldsymbol{e}_L &= \frac{\boldsymbol{\omega} \times \boldsymbol{u}}{\|\boldsymbol{\omega} \times \boldsymbol{u}\|}, \qquad \sin\theta = \frac{\|\boldsymbol{\omega} \times \boldsymbol{u}\|}{\omega v}. \end{aligned} \tag{6}$$

Lift is zero for zero spin, zero airspeed or parallel spin and airflow.

1. **Dynamic-pressure and area scale $\rho A v^2/2$.** Density, projected area and speed have the same meanings as in the drag force.

2. **Magnus direction e_L.** This unit vector is perpendicular to airflow. Topspin bends the path downward, backspin upward, and sidespin horizontally.

3. **Angular projection sin θ.** The transverse component is one for perpendicular spin and airflow and vanishes for parallel directions. This projection extends the transverse empirical law to an oblique spin axis.

4. **Lift coefficient $C_L(Re, Sp)$.** Rotation and boundary-layer state jointly determine lift.

   Rotation creates an asymmetric flow and a transverse Magnus force. Its magnitude depends on the boundary-layer state as well as spin, so a spin-only relation does not describe all flow conditions. Kiratidis and Leinweber [16] introduce the reference power law in their Eq. (8),

   $$C_L^{\text{fit}}(Sp) = 1.15 Sp^{0.83}, \tag{7}$$

   and scale it using drag as an indicator of flow state in their Eq. (9) [16]. We use that complete relation:

   $$C_L(Re, Sp) = C_L^{\text{fit}}(Sp) \frac{C_{D,\text{ceil}} - \min[C_D(Re, Sp), C_{D,\text{ceil}}]}{C_{D,\text{ceil}} - C_{D,\text{ref}}}, \tag{8}$$

   with the Teamgeist reference quantities

   $$C_{D,\text{ceil}} = f(0), \qquad C_{D,\text{ref}} = C_D(333793, 0.19). \tag{9}$$

   Here $C_{D,\text{ceil}} \simeq 0.4927$ is the zero-spin low-speed ceiling. The reference drag is evaluated with Eq. (5), including its spin term and using the same R and ν as the calculation. At the reference $(Re, Sp) = (333793, 0.19)$ the scaling factor is one and $C_L \simeq 0.290$. The ceiling, reference condition and use of drag to reduce lift follow Kiratidis and Leinweber [16]; Eq. (8) supplies the final lift coefficient in the force model.

5. **Lift correction $\alpha_L \geq 0$.** This dimensionless multiplier is one in forward prediction and is fitted to early positions when inverse prediction includes lift.

Comparison with published wind-tunnel data. Figure 4 uses the same Asai experiment and flow condition. The complete relation reproduces increasing lift with Sp and lies above all five measured values: MAE 0.04505, RMSE 0.05040 and maximum difference 0.07325. This larger discrepancy motivates the inverse lift correction α_L.

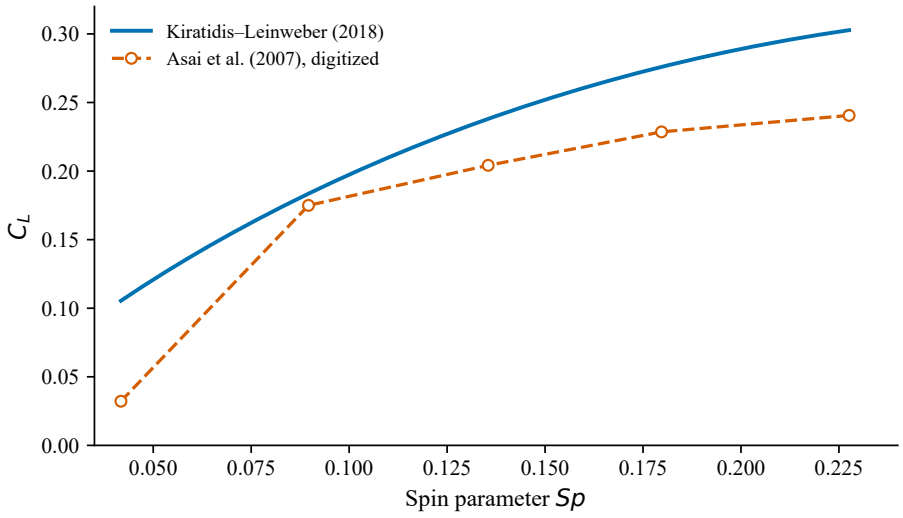


Figure 4: Teamgeist transverse lift at $Re = 455172$: the complete empirical relation [16] and five digitized Asai side-force measurements [15], Figure 9.

3.3 Dynamical Equations

Newton's second law and angular-momentum balance give

$$m\dot{\boldsymbol{v}} = \sum \boldsymbol{F}, \qquad \frac{d\boldsymbol{L}}{dt} = \boldsymbol{Q}, \qquad \boldsymbol{L} = I_b\boldsymbol{\omega}. \tag{10}$$

With zero applied torque and constant spherical inertia, the main model is

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ \dot{\boldsymbol{\omega}} = \boldsymbol{0}. \end{cases} \tag{11}$$

Its initial conditions are

$$\begin{cases} \boldsymbol{r}(0) = \boldsymbol{r}_0, \\ \boldsymbol{v}(0) = (v_{x0}, v_{y0}, v_{z0}), \\ \boldsymbol{\omega}(0) = (\omega_{x0}, \omega_{y0}, \omega_{z0}). \end{cases} \tag{12}$$

The observed initial position fixes the origin of the flight. The six velocity and spin components determine its initial motion. In still air, drag removes mechanical energy while Magnus lift does no work along the velocity; gravity converts kinetic and potential energy during flight.

4 Model Solving

Forward prediction explains how a prescribed kick develops under different force assumptions. Inverse prediction then uses early positions from three actual penalties to estimate motion and predict the continuation. Both calculations use Eq. (11) and the same four-model hierarchy.

Table 3: Four-model hierarchy and actual inverse quantities. All models include gravity and constant spin.

Model	Drag	Magnus lift	Inverse quantities
M1	none	none	$\boldsymbol{v}_0$ (3)
M2	constant C_D^*	none	$\boldsymbol{v}_0, \alpha_D$ (4)
M3	constant C_D^*	constant C_L^*	$\boldsymbol{v}_0$, spin axis, α_D, α_L (7)
M4	$C_D(Re, Sp)$	$C_L(Re, Sp)$	$\boldsymbol{v}_0, \boldsymbol{\omega}_0, \alpha_D, \alpha_L$ (8)

The constant coefficients use the published reference point:

$$C_D^* = C_D(333793, 0.19), \qquad C_L^* = C_L(333793, 0.19) = 1.15(0.19)^{0.83}. \tag{13}$$

For the reference ball and air, $C_D^* = 0.2415$ and $C_L^* = 0.2898$. M3 uses a unit vector to represent spin direction; its constant coefficient makes the spin magnitude absent from the force. M4 uses the full spin vector.

4.1 Forward Prediction

4.1.1 Prescribed conditions and solution

Five kick situations are represented by six initial states in Table 4. These are chosen numerical examples: power, placement, sidespin, a paired topspin/backspin comparison, and a chip. Each begins at $\boldsymbol{r}_0 = (0, 0, 0.11)$ m with

$$\boldsymbol{v}_0 = v_0(\cos\beta, 0, \sin\beta), \qquad \alpha_D = \alpha_L = 1. \tag{14}$$

The top- and backspin pair has the same speed and launch angle, isolating the direction of rotation. For every state, all four models are integrated by adaptive Runge–Kutta (DOP853). A forward root of $x - 11$ locates the goal crossing and a descending root of $z - R$ locates ground contact. The relative and absolute integration tolerances are 10^{-10} and 10^{-12}, with a maximum step of 0.005 s. Table 5 gives the resulting endpoint and time.

Table 4: Prescribed initial conditions.

Kick	v_0 (m/s)	β (°)	$\boldsymbol{\omega}_0$ (rad/s)
Power	27	8	$(0, 0, 0)$
Placement	20	11	$(0, 0, 0)$
Curved	24	10	$(0, 0, 35)$
Topspin	24	12	$(0, 35, 0)$
Backspin	24	12	$(0, -35, 0)$
Chip	14	20	$(0, -15, 0)$

Table 5: First goal-plane or ground endpoint for all 24 flights.

Kick	Model	End	x (m)	y (m)	z (m)	t (s)
Power	M1	goal	11.000	0.000	0.826	0.4114
Power	M2	goal	11.000	0.000	0.741	0.4420
Power	M3	goal	11.000	0.000	0.741	0.4420
Power	M4	goal	11.000	0.000	0.776	0.4296
Placement	M1	goal	11.000	0.000	0.708	0.5603
Placement	M2	goal	11.000	0.000	0.551	0.6020
Placement	M3	goal	11.000	0.000	0.551	0.6020
Placement	M4	goal	11.000	0.000	0.617	0.5848
Curved	M1	goal	11.000	0.000	0.987	0.4654
Curved	M2	goal	11.000	0.000	0.879	0.5000
Curved	M3	goal	11.000	0.943	0.876	0.5027
Curved	M4	goal	11.000	0.892	0.882	0.5008
Topspin	M1	goal	11.000	0.000	1.371	0.4686
Topspin	M2	goal	11.000	0.000	1.261	0.5036
Topspin	M3	goal	11.000	0.000	0.326	0.4998
Topspin	M4	goal	11.000	0.000	0.382	0.4983
Backspin	M1	goal	11.000	0.000	1.371	0.4686
Backspin	M2	goal	11.000	0.000	1.261	0.5036
Backspin	M3	goal	11.000	0.000	2.217	0.5130
Backspin	M4	goal	11.000	0.000	2.173	0.5108
Chip	M1	goal	11.000	0.000	0.684	0.8361
Chip	M2	goal	11.000	0.000	0.328	0.8994
Chip	M3	goal	11.000	0.000	1.221	0.9156
Chip	M4	goal	11.000	0.000	0.368	0.9489

4.1.2 Case 1: Power shot

A high launch speed leaves little time for downward gravitational deflection. Drag removes forward momentum and increases the time required to reach the goal. In the zero-spin condition, M2 and M3 coincide, while M4 changes drag with the decelerating airflow.

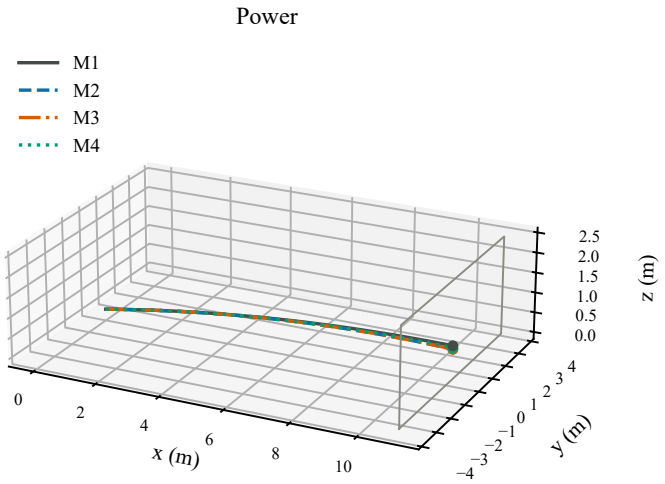


Figure 5: Power shot under the same prescribed initial condition in M1–M4. Endpoint markers denote goal crossing or ground contact.

4.1.3 Case 2: Placement shot

The lower launch speed gives gravity more time to bend the path. The comparison shows how the assumed drag coefficient changes both the travel time and crossing height. The initial direction is common to all models.

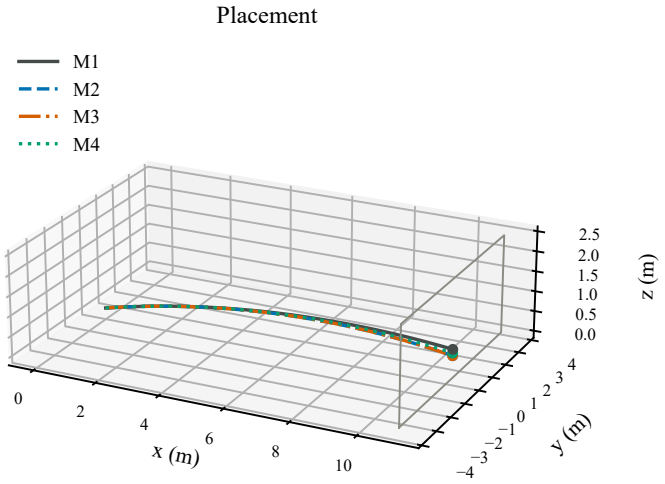


Figure 6: Placement shot with zero spin. M2 and M3 have identical forces and overlapping trajectories.

4.1.4 Case 3: Curved shot

Vertical-axis rotation creates a horizontal Magnus force. M1 and M2 preserve the initial horizontal direction; M3 and M4 develop lateral curvature. Their different coefficient descriptions change the amount of bending. The M4 goal crossing is $(0.8921, 0.8822)$ m at 0.5008 s. This case also supplies the reference state for the rotational-damping comparison in Section 6.2.

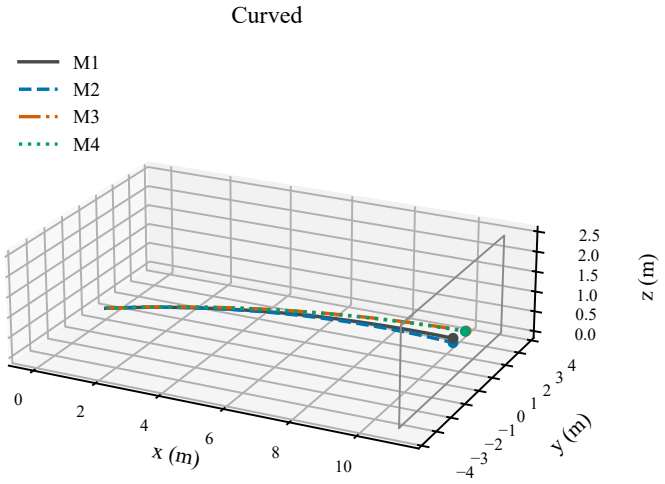


Figure 7: Sidespin redirects the ball horizontally. The four models share speed, elevation and initial spin.

4.1.5 Case 4: Topspin and backspin

Changing the sign of ω_y reverses the vertical Magnus force. Topspin increases downward curvature; backspin opposes the gravitational fall and raises the trajectory. Equal launch speed and angle make this contrast visible directly. Table 5 records each model's actual termination.

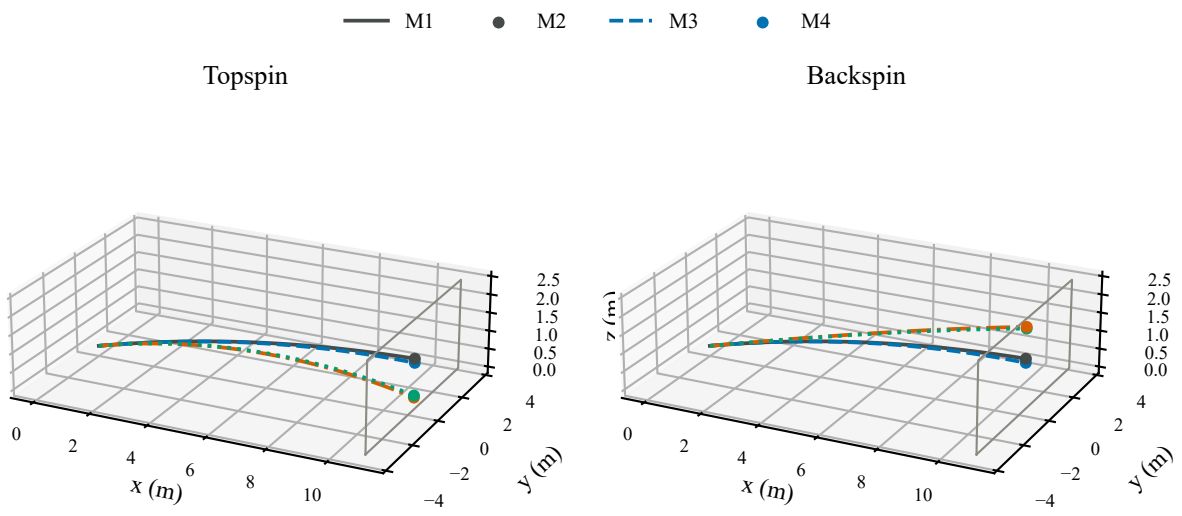


Figure 8: Matched topspin and backspin examples. Only the sign of transverse spin changes between the two panels.

4.1.6 Case 5: Chip shot

The chip combines a larger elevation angle, lower speed and backspin. The initial vertical velocity builds an arc, while gravity, drag and upward Magnus lift determine how much height remains near the goal. Its slower airflow also makes the speed-dependent drag transition more relevant.

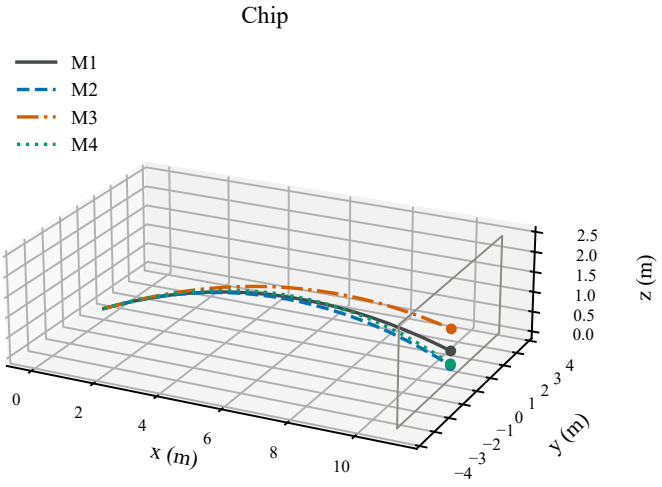


Figure 9: The prescribed chip illustrates the combined effect of launch angle, slower airflow and backspin.

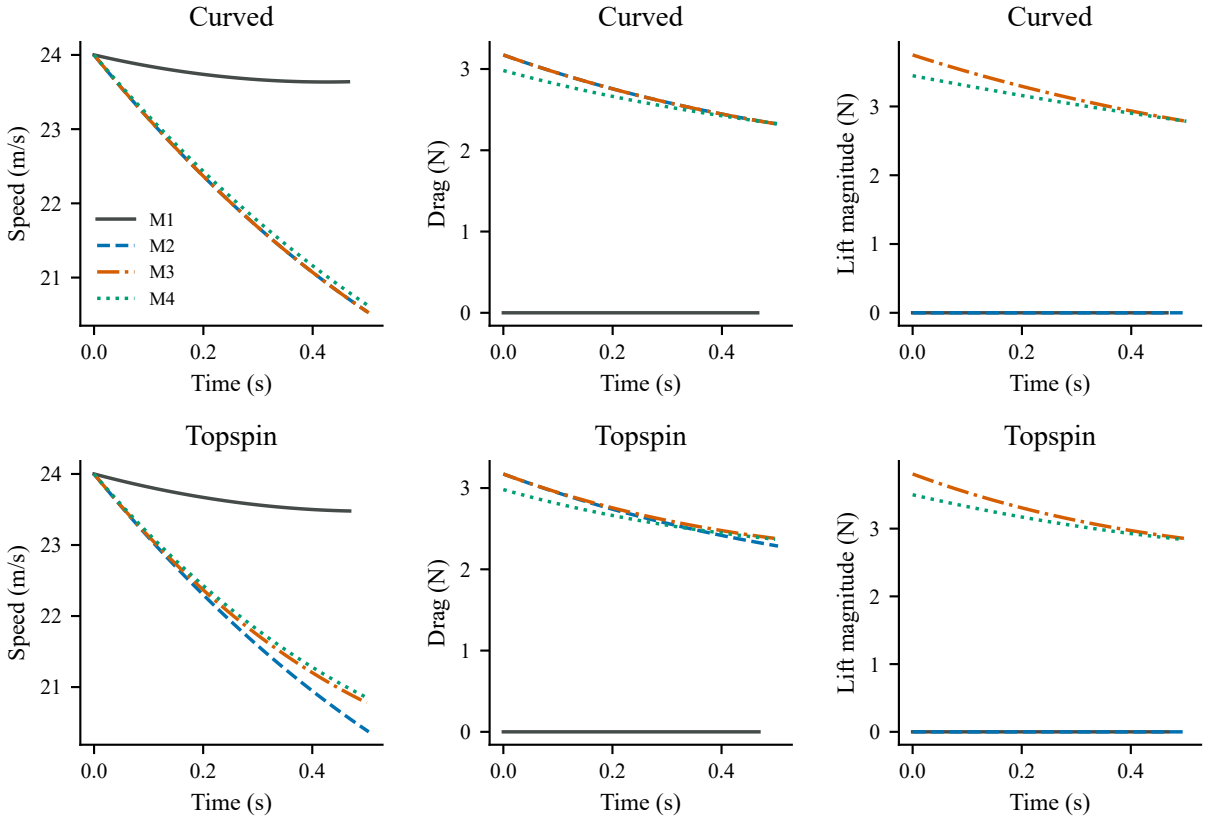


Figure 10: Speed, drag and lift magnitudes for the curved and topspin examples. Each curve ends at that model's first goal crossing or ground contact.

The force histories connect the paths to their causes. Dynamic pressure decreases as drag removes speed. A constant coefficient still gives a changing force through the v^2 factor. In M4, changing Re and Sp additionally changes the coefficients. The spin direction decides whether lift acts chiefly sideways or vertically.

4.2 Inverse Prediction from Three Real Penalties

4.2.1 Measurements, coordinate system and input times

The Sportec Open DFL data associate actual match events with TRACAB Gen5 multi-camera tracking at 25 Hz [10, 11]. We use penalties by Dejan Ljubicic (J03WMX), Dawid Igor Kownacki (J03WOH), and Philipp Klement (J03WR9). The supplied CSV retains 39 source frames, respectively 15, 12 and 12. Positions are used as published; no added observation-noise model is introduced.

The source horizontal origin is the pitch centre. All three shots travel towards $X = 52.5$ m. To use the competition's penalty-mark coordinates, we apply

$$x = X - 41.5, \qquad y = Y, \qquad z = Z_{\text{original}}, \qquad x_g = 11 \text{ m}. \tag{15}$$

The nominal penalty mark is the horizontal origin, so each actual first observation retains its measured offset. Released Z values are interpreted as metres and ball-centre height. Mass, radius and air properties use the model reference settings in Section 2; the supplied source leaves per-ball mass and radius unreported.

The first available source frame defines $t = 0$ for each fit. Table 6 lists the actual selected times: J03WMX and J03WOH use 0, 0.04 and 0.08 s; J03WR9 uses 0, 0.08 and 0.12 s because its first two records repeat. That repeated frame remains in the measured curve. Later samples supply the comparison, ending at the observed goal crossing. Original post-goal records remain in the raw CSV. Crossings between frames use linear interpolation, displayed separately from measured sample markers.

Table 6: Frame counts, fitting times and measured local starting positions.

Event	Frames	Fit times (s)	$\boldsymbol{r}_0$ (m)
J03WMX	15	0.00,0.04,0.08	$(0.71, 0.05, 0.16)$
J03WOH	12	0.00,0.04,0.08	$(0.06, -0.04, 0.11)$
J03WR9	12	0.00,0.08,0.12	$(1.47, 0.49, 0.23)$

4.2.2 Bounded inversion and consistent selection

For observed positions $\boldsymbol{p}_1, \boldsymbol{p}_2$ at t_1, t_2, the position objective is

$$J_r = \sum_{i=1}^{2} \|\boldsymbol{r}(t_i; \boldsymbol{\eta}) - \boldsymbol{p}_i\|^2, \qquad \boldsymbol{\eta}_{\text{M4}} = (\boldsymbol{v}_0, \boldsymbol{\omega}_0, \alpha_D, \alpha_L). \tag{16}$$

Next we favor force corrections near the published scale,

$$J_\alpha = (\alpha_D - 1)^2 + (\alpha_L - 1)^2, \tag{17}$$

using the multipliers present in each model. Among equivalent M4 states, the smallest spin magnitude selects a representative. The numerical objective allowances are 6×10^{-12} m^2 for J_r and 10^{-8} for J_α.

The implemented search uses the component constraints in Table 7. These are declared analysis settings. The speed and spin scales refer to the empirical and experimental ranges [16, 15]; the component boxes and the force-correction interval specify this inverse calculation. M3's unit spin axis carries direction only.

Table 7: Component bounds and force-correction settings for inversion.

Quantity	Search interval
v_{x0}	$[0, 35]$ m/s
v_{y0}, v_{z0}	$[-35, 35]$ m/s
$\omega_{x0}, \omega_{y0}, \omega_{z0}$ (M4)	$[-20\pi, 20\pi]$ rad/s
Azimuth, elevation (M3)	$[-\pi, \pi], [-\pi/2, \pi/2]$ rad
α_D, α_L	$[0.5, 1.5]$

M2 uses one displacement-based start, M3 uses six axis-direction starts, and M4 uses 19 starts including zero spin and signed coordinate-axis spins. Least squares allows 240 function evaluations per start. Coefficient and spin selection use the same candidate state when evaluating the residual and the secondary objective. The fitted state and configuration then enter the same dynamics for prediction.

4.2.3 Predicted and sampled trajectories

Figures 11–13 compare all four models with each sampled flight. Open circles identify the three fit inputs. Time plots end at the measured crossing time or an earlier model termination; squares mark ground contact and an asterisk identifies an iteration-limit candidate. The outcome table retains all 12 calculations.

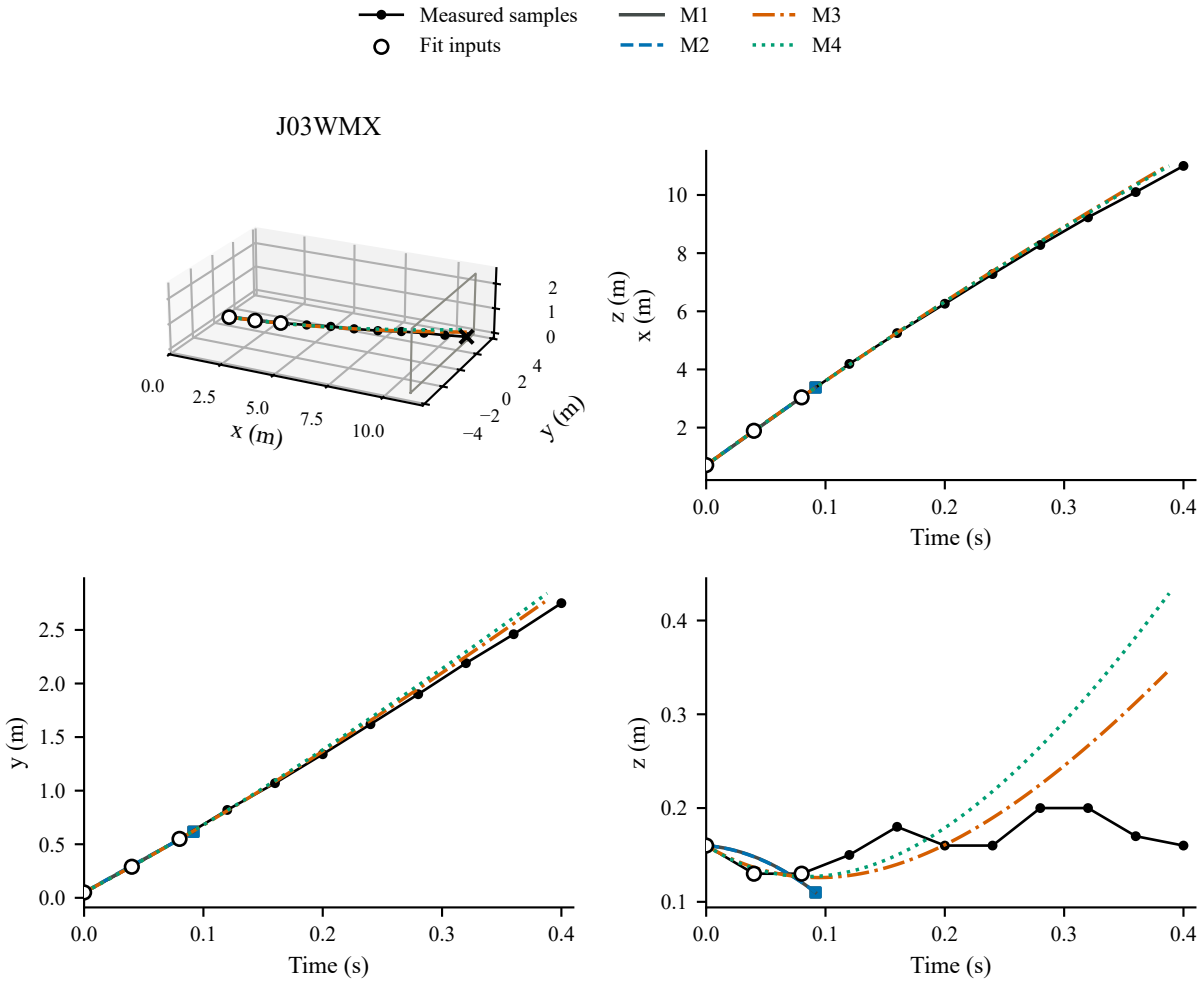


Figure 11: J03WMX: predicted spatial path and coordinate histories against the published samples.

The recorded path stays close to the turf. Gravity-only and drag-only predictions reach the ground before the goal; upward Magnus force in M3 and M4 sustains the low flight. The M4 candidate crossing error is 0.284 m, with arrival-time difference -0.012 s.

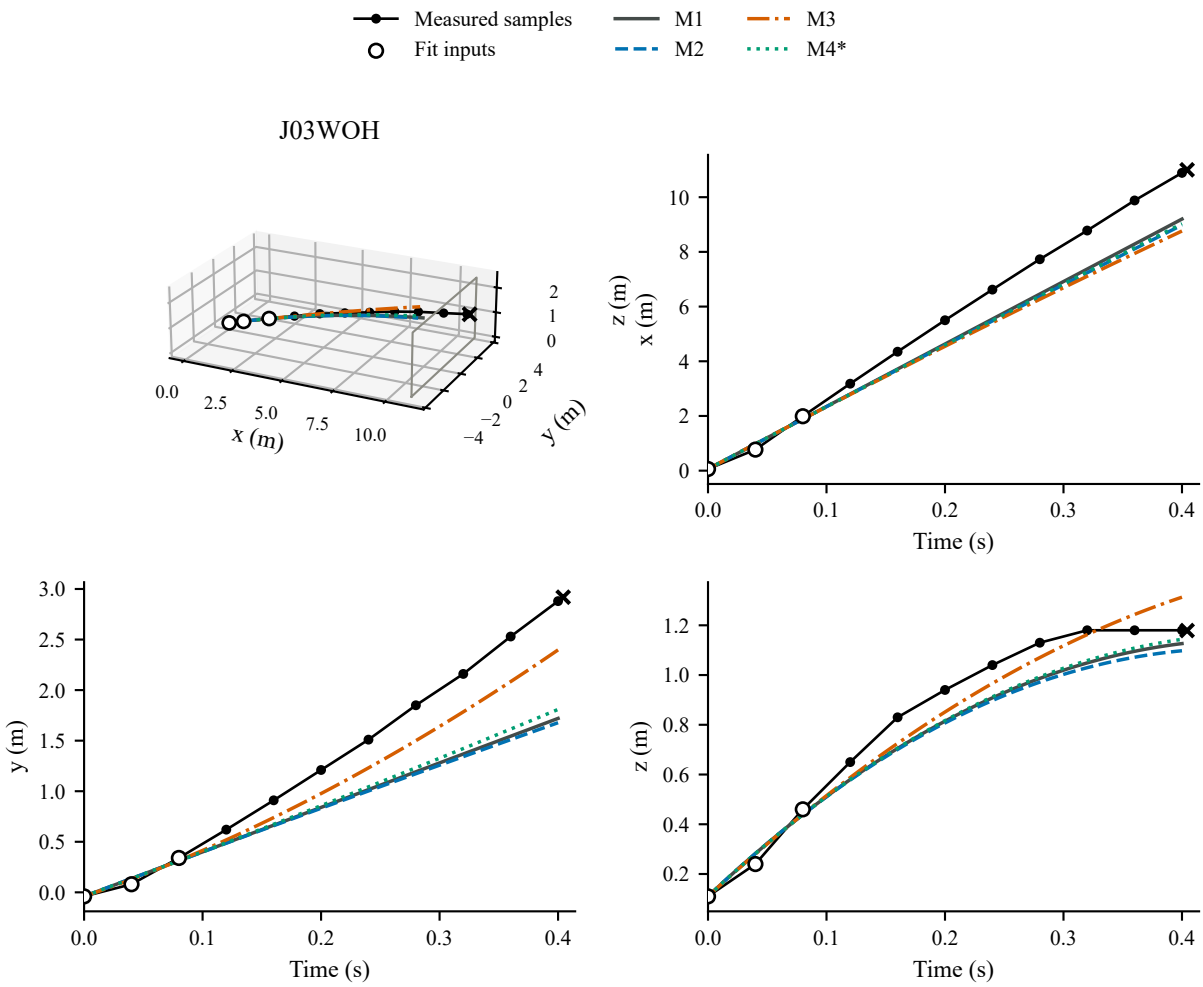


Figure 12: J03WOH: the recorded first displacement is shorter than the next; the comparison uses the original selected frames.

The first two horizontal increments are 0.71 and 1.22 m in equal time intervals. This accelerating early sequence drives the drag correction to its lower bound. The M4 search reaches the iteration limit; its candidate is retained and marked in the figure and table. The M4 candidate crossing error is 0.664 m, with arrival-time difference +0.086 s.

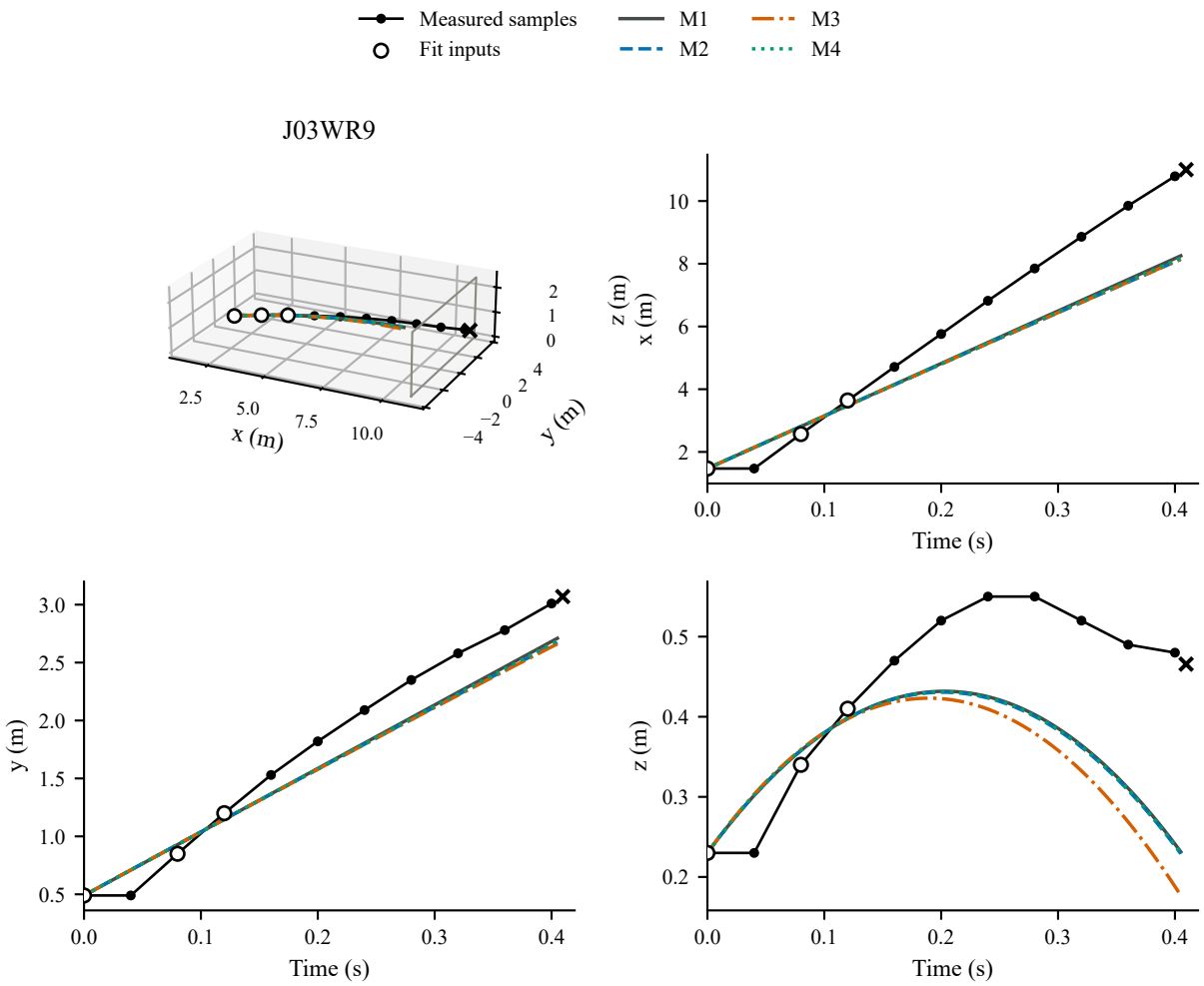


Figure 13: J03WR9: the repeated 0.04 s record remains in the sampled curve, while fitting uses 0, 0.08 and 0.12 s.

Using 0, 0.08 and 0.12 s avoids treating the duplicated first position as a separate displacement. The measured lateral motion is substantial, and fitted lift direction controls the later sideways bending.

Table 8: All inversions: maximum early coordinate residual, later 3D RMSE/sample count, goal error and arrival-time error. Limit rows describe the saved iteration-limit candidate.

Event	Model	Status	Resid. (m)	RMSE (m)/n	Goal (m)	Δt (s)
WMX	M1	ground	1.83×10^{-2}	—/0	—	—
WMX	M2	ground	1.81×10^{-2}	—/0	—	—
WMX	M3	goal	6.74×10^{-3}	0.161/7	0.185	-0.014
WMX	M4	goal	5.84×10^{-3}	0.176/7	0.284	-0.012
WOH	M1	goal	2.04×10^{-1}	1.387/8	0.853	0.074
WOH	M2	goal	2.06×10^{-1}	1.500/8	0.856	0.090
WOH	M3	goal	2.08×10^{-1}	1.463/8	0.505	0.107
WOH	M4	limit	2.06×10^{-1}	1.441/8	0.664	0.086
WR9	M1	ground	2.40×10^{-1}	1.816/7	—	—
WR9	M2	ground	2.42×10^{-1}	1.879/7	—	—
WR9	M3	ground	2.42×10^{-1}	1.880/7	—	—
WR9	M4	ground	2.41×10^{-1}	1.855/7	—	—

The later-position RMSE is computed only at held-out source times within both the observed pre-goal flight and the model's valid flight interval; its sample count is listed. Goal error compares (y, z) at the model's own crossing, and time error compares crossing times. These quantities distinguish early-coordinate reproduction, continuation of the path, and the final interception geometry.

5 Sensitivity Analysis

To be completed.

6 Discussion of Other Factors

The main model isolates gravity, drag and Magnus lift. We now examine the physical effects omitted from its geometry, spin evolution and ambient airflow.

6.1 Asymmetric Surface Geometry

Paudel and Yan [13] performed full-flight flow simulations using realistic football geometry and a smooth-sphere comparison. For launch speeds of 10, 15 and 20 m/s and spin rates of 31.5, 63 and 94.5 rad/s, their results show that seams and surface structure alter separation, wake dynamics and Magnus force. The smooth sphere produces greater lateral bending in their nine comparisons, and vertical flight-time differences become more visible at higher speed and rotation.

The spherical geometry in the present report describes the centre-of-mass motion and projected area. Its force coefficients come from football measurements, so the average surface influence enters through C_D and C_L. Explicit panel orientation and asymmetric wake fluctuations provide a further level of description when lateral-force details are important. These published results support retaining football-specific coefficients in a spherical trajectory model.

6.2 Rotation and Aerodynamic Damping

Rotation changes Magnus force, so a declining spin can change the continuation of a curved shot. According to Grad et al. [17], Supplementary Section 8.4, and the MuJoCo ellipsoid model [18], the spherical inertial and viscous damping torques are

$$\boldsymbol{Q} = -k_2\omega\boldsymbol{\omega} - k_1\boldsymbol{\omega}, \quad k_2 = \frac{8\pi}{15}\rho C_{\text{angular}} R^5, \quad k_1 = 8\pi\mu R^3. \tag{18}$$

The quadratic term describes high-Reynolds-number angular drag; the linear term is viscous resistance. We use MuJoCo's documented reference value $C_{\text{angular}} = 1.5$ [18]. Defining $q = \rho v^2/2$ and $C_{Q,v} = \|\boldsymbol{Q}\|/(qAR)$ gives

$$C_{Q,v} = \frac{16}{15} C_{\text{angular}} Sp^2 + \frac{32Sp}{Re}, \qquad C_{Q,\omega} = \frac{C_{Q,v}}{Sp^2} = \frac{16}{15} C_{\text{angular}} + \frac{32}{Re\, Sp}. \tag{19}$$

$C_{Q,\omega}$ uses rotating surface speed $R\omega$ in its reference pressure. The factor $1/Sp^2$ converts between the two normalizations. The viscous-to-inertial torque ratio is

$$\frac{Q_{\text{viscous}}}{Q_{\text{inertial}}} = \frac{30}{C_{\text{angular}} Re\, Sp} = \frac{15\nu}{C_{\text{angular}} R^2 \omega}. \tag{20}$$

It is 3.54×10^{-4} at the prescribed curved-shot launch, so the quadratic term dominates. Following the thin-shell inertia used by Maponi and Piombin [12], $I_b = 2mR^2/3$. Adding the high-Reynolds-number torque gives

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ I_b\dot{\boldsymbol{\omega}} = -k_2\omega\boldsymbol{\omega}. \end{cases} \tag{21}$$

The torque opposes spin and removes rotational energy. With this isolated quadratic law, $\omega(t) = \omega_0/[1 + (k_2/I_b)\omega_0 t]$, providing an integration check.

Use the curved shot of Section 4.1, with initial velocity $(23.6354, 0, 4.1676)$ m/s, spin $(0, 0, 35)$ rad/s and unit force corrections. Holding this state fixed, we compare constant spin with Eq. (21).

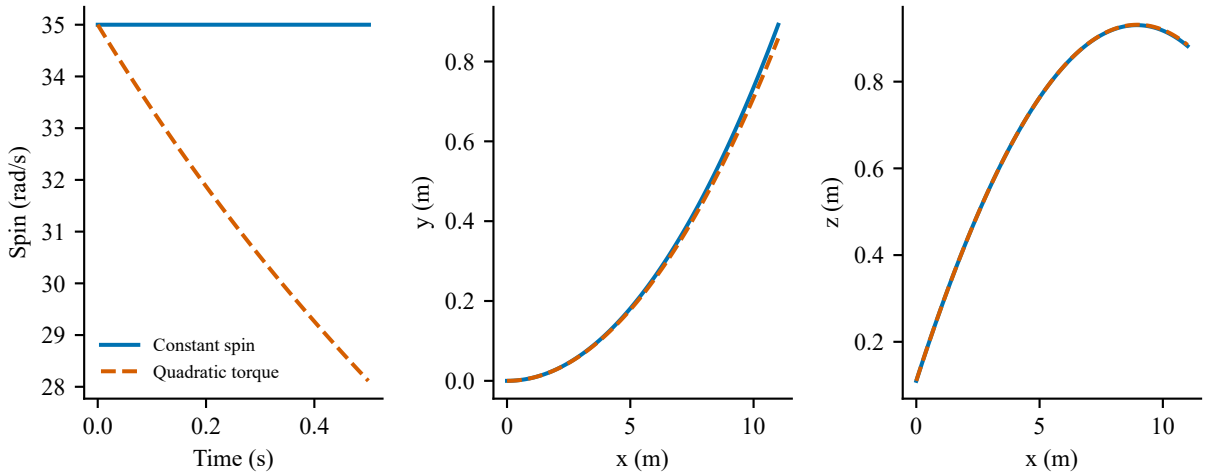


Figure 14: Same-initial-state comparison of constant spin and high-Reynolds-number rotational damping for the prescribed curved shot. The documented angular-drag reference is 1.5, and the ball uses thin-shell inertia.

The damped crossing is $(0.8579, 0.8848)$ m, compared with $(0.8921, 0.8822)$ m at constant spin. The displacement is 0.0343 m (0.312% of 11 m). Spin falls by 19.671%, and crossing time changes by -0.233%. Adding viscous torque shifts the damped crossing by 1.2e-05 m. The short-flight endpoint is less sensitive than spin magnitude to this reference torque.

6.3 Wind Speed and Force Estimates

Wind changes the airflow through $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$ [12]. At fixed coefficients, a wind speed w along the flight direction changes the dynamic-pressure scale by

$$\frac{q_{\text{head}}}{q_0} = (1 + w/v)^2, \qquad \frac{q_{\text{tail}}}{q_0} = (1 - w/v)^2, \qquad \frac{q_{\text{cross}}}{q_0} = 1 + (w/v)^2. \tag{22}$$

Crosswind also rotates the airflow by $\arctan(w/v)$, changing the force direction. Table 9 evaluates these rough estimates at the curved-shot launch speed. The listed wind speeds are illustrative conditions for comparison.

Table 9: Illustrative wind at the 24 m/s curved-shot launch: dynamic-pressure changes at fixed coefficients.

w (m/s)	Headwind (%)	Tailwind (%)	Crosswind (%)	Angle (°)
1	+8.51	-8.16	+0.17	2.39
3	+26.56	-23.44	+1.56	7.13
5	+46.01	-37.33	+4.34	11.77

A 5% limit on the headwind dynamic-pressure change corresponds to $w/v \leq \sqrt{1.05} - 1$, or $w \leq 0.59$ m/s at this launch. The still-air approximation is most suitable for indoor or weak-wind flight; stronger wind should enter through the relative-velocity calculation.

7 Conclusions

The model separates the mechanisms of gravity, drag and Magnus lift. Gravity produces downward curvature, drag reduces speed, and spin redirects the path. The complete Teamgeist relations connect coefficient changes to speed and rotation; the matched wind-tunnel comparison reproduces drag more closely than lift.

Five prescribed kick situations provide six initial states and 24 forward flights. The matched top- and backspin pair demonstrates the reversal of vertical Magnus force, while sidespin produces horizontal curvature. In the curved M4 example the goal crossing is $(0.8921, 0.8822)$ m at 0.5008 s. The same initial state is used to quantify angular damping, giving a crossing displacement of 0.0343 m under the cited reference setting.

Three actual penalties supply distinct early sampling sequences and held-out trajectories. Coordinate conversion places all three in the competition geometry. The 12 inverse outcomes report early residuals, continuation errors, goal crossings and ground termination separately, connecting the recorded motion to each force assumption. Section 5 remains reserved for sensitivity analysis and crossing-location distributions.

8 Model Evaluation

8.1 Physical Interpretation and Reproduction

The four-model hierarchy makes the role of each force visible under shared initial conditions. Forward and inverse calculations use the same state derivative and coefficient definitions. Original source coordinates, selected frames, transformed coordinates, numerical settings and candidate solutions are saved with the results.

8.2 Prediction and Further Development

The real-penalty comparison evaluates how early motion carries into the later path. Bound-active solutions and residuals identify the influence of the chosen coefficient scale and initial-state search. Ball-specific coefficients, direct spin measurements and higher-rate synchronized tracking would improve these connections. Rotational damping and wind can be included when their effect is relevant to the flight considered.

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Reproduction Code

The project contains the original penalty CSV, coordinate conversion, shared inversion, prediction results and figure scripts. The excerpts show the common state equation and force evaluation. Paper and program use the same coefficients and bounds; computation, plotting and bilingual table generation are separate commands.

```
--
def state_derivative(t, state, config):
    """The single nine-state RHS used for prediction and inverse fitting."""
    f = forces(state[3:6], state[6:9], config)
    return np.r_[state[3:6], sum(f.values())/config.mass,
                rotational_torque(state[6:9], config)/config.moment_of_inertia]
```

```
--
def forces(velocity, spin, config):
    """Return force components on ball, in newtons."""
    v = np.asarray(velocity, dtype=float) - np.asarray(config.wind, dtype=float)
    omega = np.asarray(spin, dtype=float)
    speed = np.linalg.norm(v)
    weight = np.array([0., 0., -config.mass * config.gravity if config.use_gravity else
        0.])
    drag = np.zeros(3)
    magnus = np.zeros(3)
    if speed > 1e-12:
        cd, cl, _, _, _ = coefficients(velocity, spin, config)
        q_area = 0.5 * config.air_density * np.pi * config.radius**2 * speed**2
        if config.use_drag:
            drag = -q_area * config.drag_scale * cd * v / speed
        cross = np.cross(omega, v)
        norm = np.linalg.norm(cross)
        if config.use_magnus and norm > 1e-12:
            # The literature fit is for transverse spin. Project its effect
            # onto the plane perpendicular to airflow for an oblique spin axis.
            sin_angle = norm / (np.linalg.norm(omega) * speed)
            magnus = q_area * config.lift_scale * cl * sin_angle * cross / norm
    return {"gravity": weight, "drag": drag, "magnus": magnus}
```